TENSORING A PRO-ISOMORPHISM TRIANGLE WITH A COMPLEX AND TAKING DERIVED LIMITS IS EXACT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- There is a functorial construction of tensoring with limits, giving exact functors of triangulated categories Delta: D(A) -> D(N,A), tensor, and Rlim. invoked at proof lines 23931-23936. Provides the construction whose output applied to the input triangle yields the asserted canonical triangle.

Source reference. Original remark 091J: sources/stacks-more-algebra/pinned-original-source.tex, source lines 23886–23886. Cited at statement line 23927 and again at proof lines 23931-23932 for the functorial construction whose exactness yields the asserted triangle; the construction is used inside the counted span.

Lemma 0.2. *Let A be a ring. Let $E \to D \to F \to E[1]$ be a distinguished triangle of $D(\mathbf{N}, A)$. Let (E_n), resp. (D_n), resp. (F_n) be the system of objects of $D(A)$ associated to E, resp. D, resp. F. Then for every $K \in D(A)$ there is a canonical distinguished triangle*

$$R\lim(K \otimes^{\mathbf{L}}_A E_n) \to R\lim(K \otimes^{\mathbf{L}}_A D_n) \to R\lim(K \otimes^{\mathbf{L}}_A F_n) \to R\lim(K \otimes^{\mathbf{L}}_A E_n)[1]$$

in $D(A)$ with notation as in Remark 091J.

Proof. This is clear from the construction in Remark 091J and the fact that $\Delta : D(A) \to D(\mathbf{N}, A)$, $- \otimes^{\mathbf{L}} -$, and $R\lim$ are exact functors of triangulated categories. □

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